Section 9 Scenario A natural images

BSDS500 real-image comparison with validation-tuned classical baselines

Five BSDS500 test crops. Clean qualitative panels. Metrics over SNR levels clean, 20 dB, 10 dB, and 5 dB.

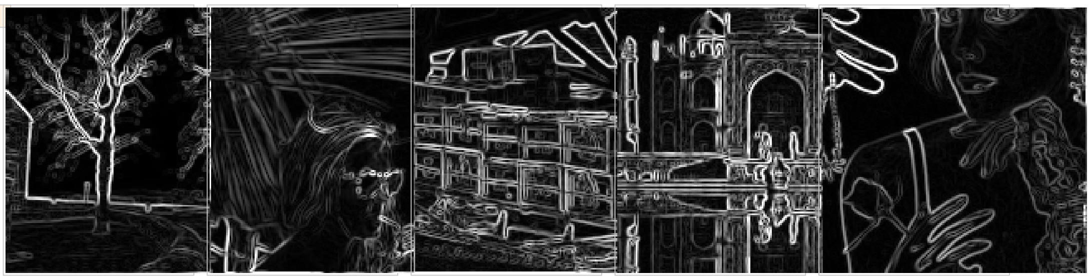
Gradient magnitude, clean real images

Method	92014-1	81090-2	188025-3	288024-4	198087-5

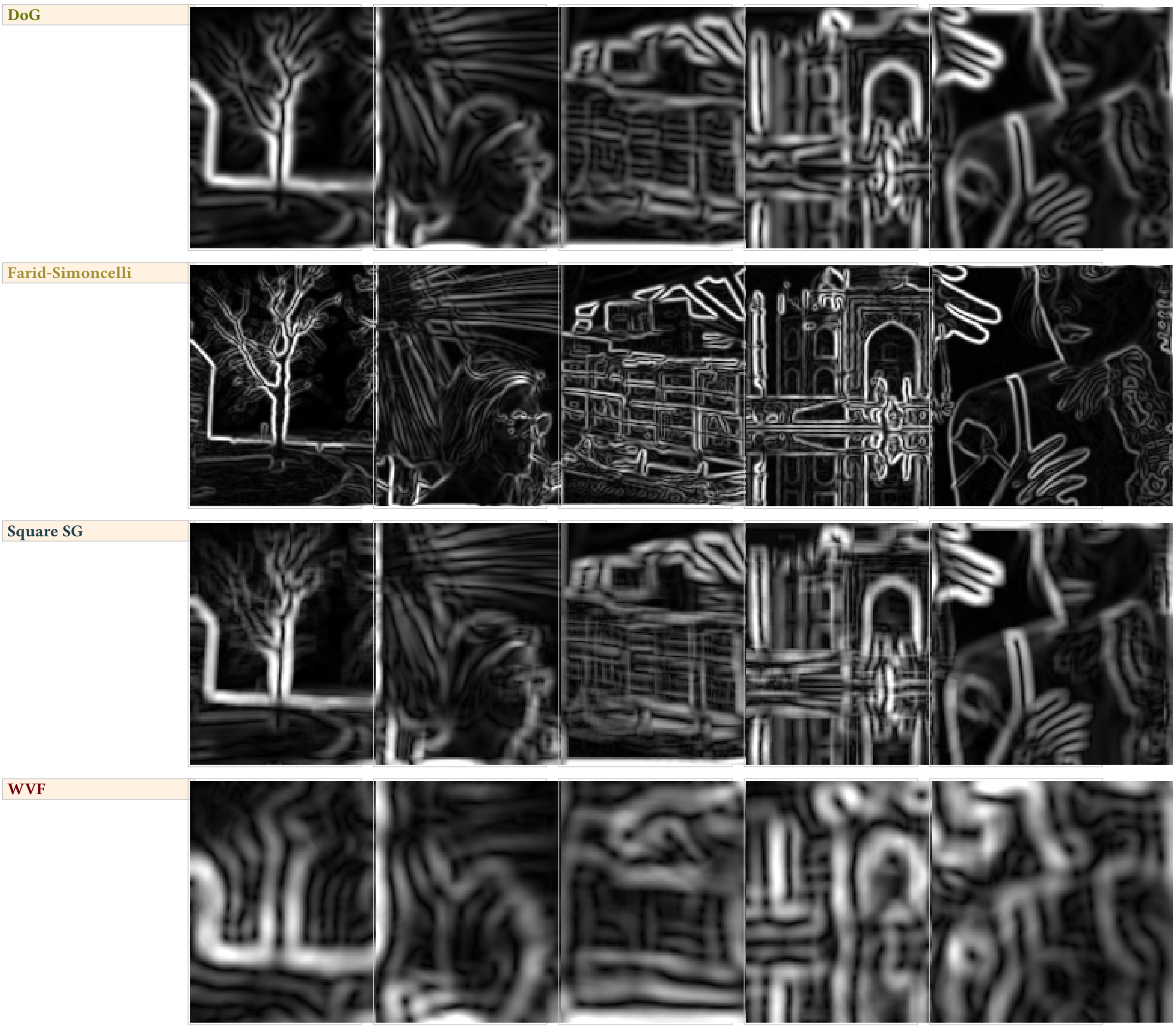
Roberts

Prewitt

Sobel



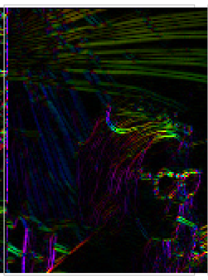
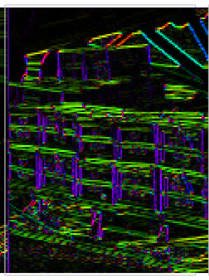
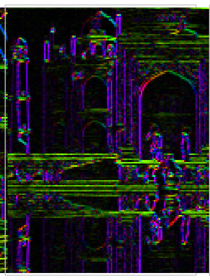
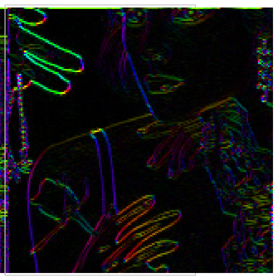
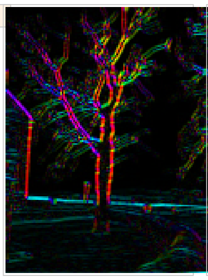
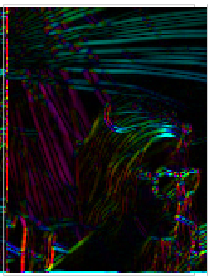
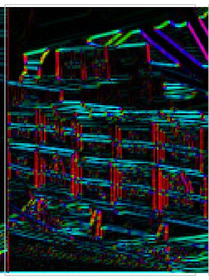
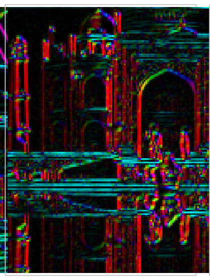
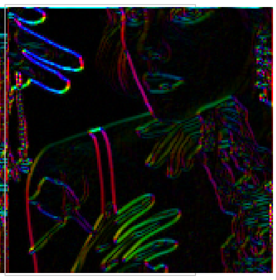
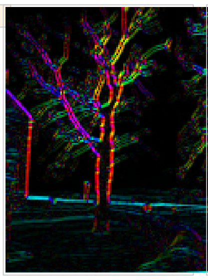
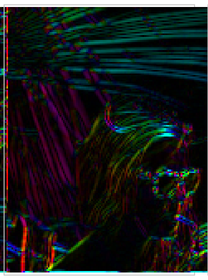
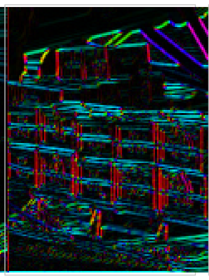
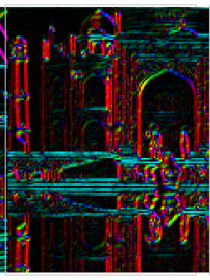
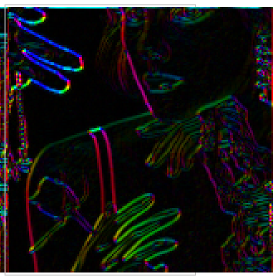
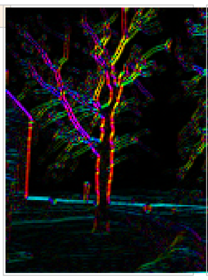
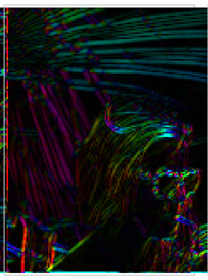
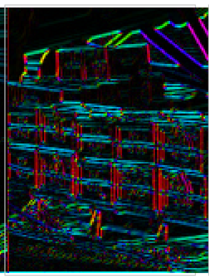
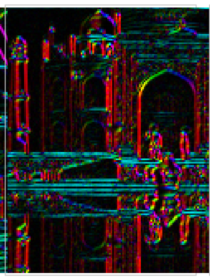
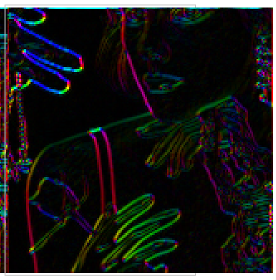
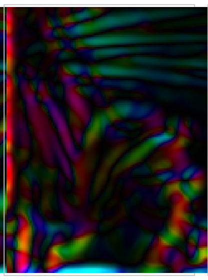
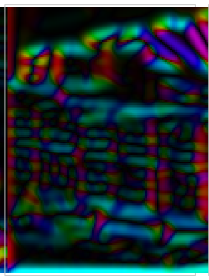
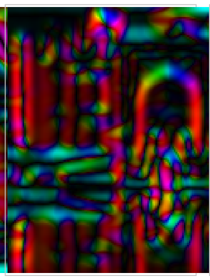
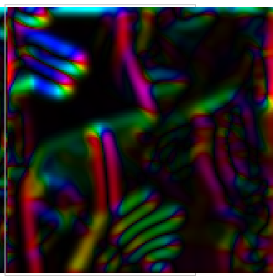
Scharr


DoG
Farid-Simoncelli
Square SG
WVF

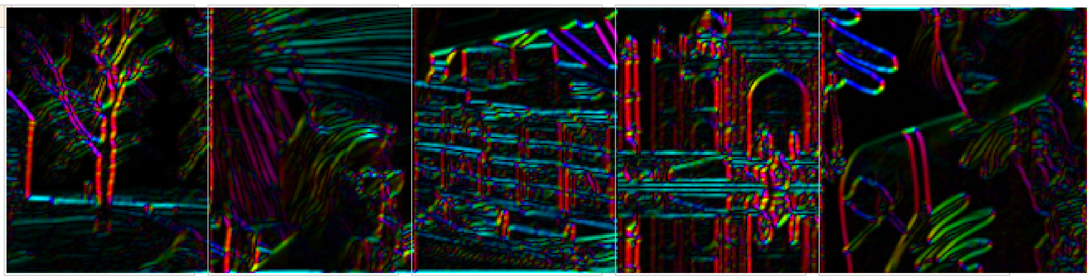
Section 9 Scenario A natural images

Orientation maps. Hue encodes edge angle modulo π and brightness follows normalized magnitude.

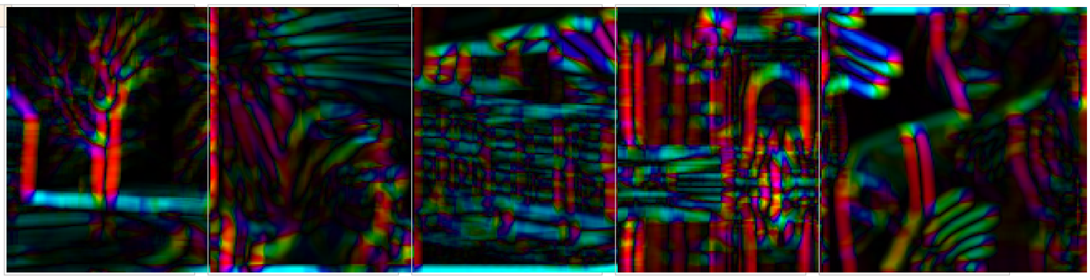
Gradient orientation, clean real images

Method	92014-1	81090-2	188025-3	288024-4	198087-5
Roberts					
Prewitt					
Sobel					
Scharr					
DoG					

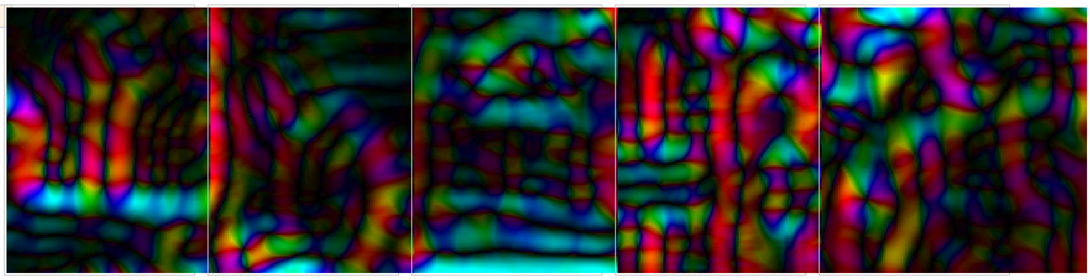
Farid-Simoncelli



Square SG



WVF



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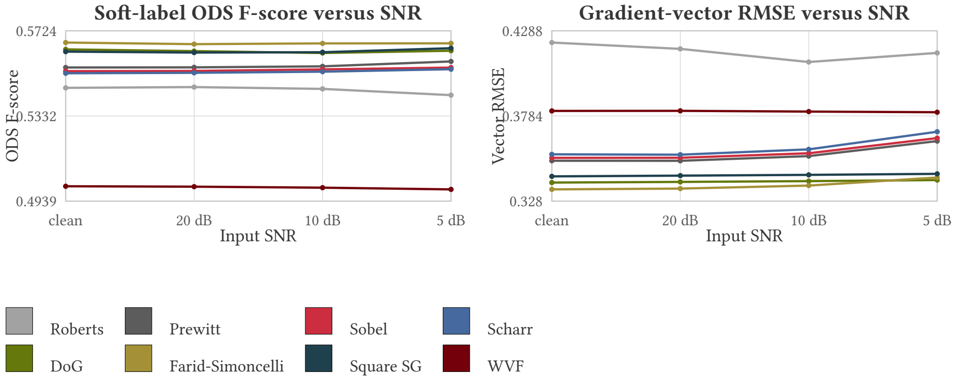
Quantitative summary across 5 crops. Noisy metrics average over 10 AWGN draws per crop.



Metric	Roberts	Prewitt	Sobel	Scharr	DoG	Farid-Simoncelli	Square SG	WVF
ODS clean	0.5461	0.5555	0.5539	0.5529	0.5639	0.567	0.5628	0.5008
ODS 10 dB	0.5457	0.5561	0.5547	0.5536	0.5622	0.5666	0.5626	0.5001
RMSE clean	0.4218	0.3519	0.3536	0.3557	0.339	0.3349	0.3426	0.3814
RMSE 10 dB	0.4103	0.3546	0.3563	0.3586	0.3398	0.3372	0.3435	0.381

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WVF radius-trace follow-up on the fixed BSDS crop set.

Garnet trace points sweep WVF across $(r, d) = (3, 5), (5, 9), (9, 11), (15, 11), (25, 11), (50, 11)$. Thin horizontal lines are the fixed baseline methods. Outlined WVF points beat the best fixed baseline at that metric and SNR.